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Guidance

# Safety Bulletin 26 – $CO_2$ fire suppression installation testing

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Contents

# 1. Summary of issue

Following a fire in the auxiliary engine room on board the Finnmaster ro-ro cargo vessel in Hull on 19 September 2021 the Marine Accident Investigation Branch (MAIB) Safety Bulletin 01/22 was issued which identified a blockage in the CO2 fire extinguishing systems pilot hose. This blockage resulted in the CO2 system failing to activate fully with only half of the intended CO2 cylinders being discharged resulting in the crew having to enter the space in breathing apparatus to fully extinguish the fire.

The blockage has been found to be the result of a faulty hose coupling in the pilot line system.

The crew were not familiar with the manual activation procedure for the CO2 system and as a result were not able to bypass the remote actuation system where the blockage had been present.

# 2. Actions to take

The MAIB have taken action to identify the source of the faulty hose couplings and identified companies that have potentially been supplied with faulty hose assemblies to recommend immediate remedial action to ensure these faulty assemblies are not fitted. The MAIB are investigating the company that supplied the hose assemblies and will take appropriate action as required.

All owners of vessels with a CO2 fire extinguishing system or other fire extinguishing system operated through gas activated valves should ensure that both a pressure test and blow through test is completed after installation or modification to the system. These tests should be completed for the main circuit delivering the extinguishing medium as well as the remote-control pilot lines to ensure they hold pressure and are not blocked in anyway.

Where there is no record of a pressure or blow-through test being conducted post installation/modification the vessel owner should undertake take these tests and the next suitable maintenance period.

Where the fire extinguishing system can be operated through a remote pilot line actuation as well as a manual activation the crew should be familiar with both means of activation. Operating the system through a manual bypass should be included in fire drills to ensure it is understood by the crew.

# 3. Further information

Further information regarding the indecent on board the Finnmaster can be found in “MAIB Safety Bulletin 1/2022 (https://www.gov.uk/maib-reports/safety-warning-issued-after-discovery-of-blocked-fixed-co2-fire-extinguishing-system-pilot-hoses) Blockage of fixed CO2 fire extinguishing system pilot hoses identified following a fire on board the roll-on/roll-off cargo ship Finnmaster in Hull, England on 19 September 2021”.

Details on MCA test requirements for CO2 fire extinguishing systems can be found in MSIS 12 Chapter 7 (https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/870369/msis012ch7rev1012.pdf) Section 7.3.3

Any queries should be directed to the MCA at:

Email: marine.technology@mcga.gov.uk

Tel: +44 (0) 203 817200

> Maritime and Coastguard Agency,
> Bay 2/23,
> Spring Place,
> 105 Commercial Road,
> Southampton,
> SO15 1EG.